Before leaving the topic of computed tomography, it should be mentioned that there are several similar imaging techniques in the medical field. All use extensive amounts of DSP. **Positron emission tomography (PET)** involves injecting the patient with a mildly radioactive compound that emits *positrons*. Immediately after emission, the positron annihilates with an electron, creating two gamma rays that exit the body in exactly opposite directions. Radiation detectors placed around the patient look for these back-to-back gamma rays, identifying the location of the *line* that the gamma rays traveled along. Since the point where the gamma rays were created must be somewhere along this line, a reconstruction algorithm similar to computed tomography can be used. This results in an image that looks similar to CT, except that *brightness* is related to the amount of the radioactive material present at each location. A unique advantage of PET is that the radioactive compounds can be attached to various substances used by the body in some manner, such as glucose. The reconstructed image is then related to the concentration of this biological substance. This allows the imaging of the body's *physiology* rather than simple *anatomy*. For example, images can be produced showing which portions of the human brain are involved in various mental tasks.

A more direct competitor to computed tomography is **magnetic resonance imaging (MRI)**, which is now found in most major hospitals. This technique was originally developed under the name **nuclear magnetic resonance (NMR)**. The name change was for public relations when local governments protested the use of anything *nuclear* in their communities. It was often an impossible task to educate the public that the term *nuclear* simply referred to the fact that all atoms contain a *nucleus*. An MRI scan is conducted by placing the patient in the center of a powerful magnet. Radio waves in conjunction with the magnetic field cause selected nuclei in the body to resonate, resulting in the emission of secondary radio waves. These secondary radio waves are digitized and form the data set used in the MRI reconstruction algorithms. The result is a set of images that appear very similar to computed tomography. The advantages of MRI are numerous: good soft tissue discrimination, flexible slice selection, and not using potentially dangerous x-ray radiation. On the negative side, MRI is a more expensive technique than CT, and poor for imaging bones and other hard tissues. CT and MRI will be the mainstays of medical imaging for many years to come.